Lab 2 — Responsive Navbar and Grid Drill

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Course	CSC336 Web Technologies
Folder	`lab-2/`

1. Task statement

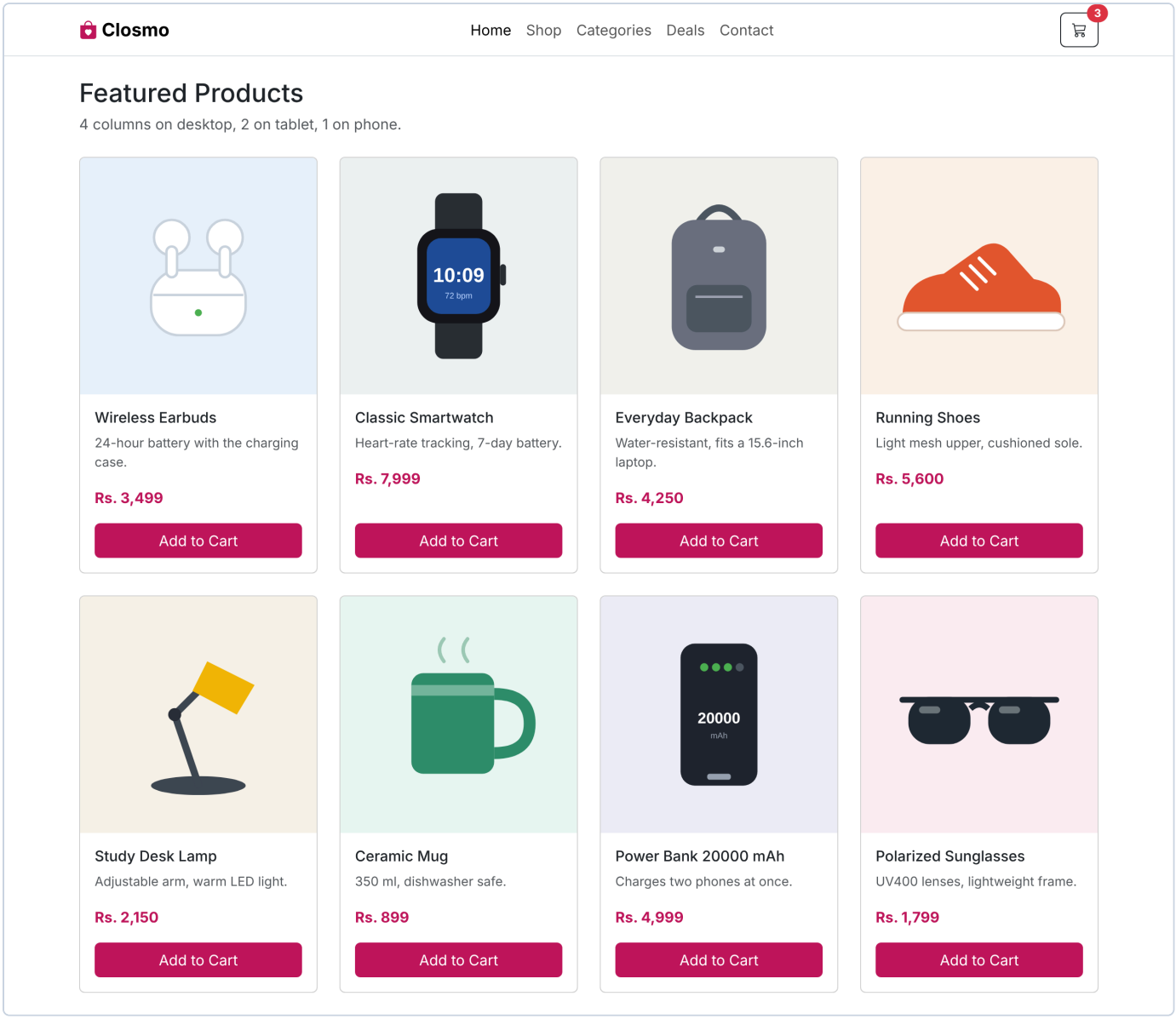
Lab 2 — Responsive Navbar and Grid Drill
Difficulty: Easy–Medium · Covers: Lectures 7–10 · Folder: `lab-2/`

A focused, smaller-scale rehearsal for Assignment 2: build just a responsive navbar (logo left, links center, cart icon right, collapsing into a hamburger menu below 768px) and a 4-column-desktop / 2-column-tablet / 1-column-phone product grid, using Flexbox or Grid plus Bootstrap or Tailwind. Content can be placeholder text/images — the point is the responsive mechanics, not the final design.

Deliverables in `lab-2/`: `index.html`, your CSS/framework files, and your completion PDF (with screenshots at phone/tablet/desktop widths).

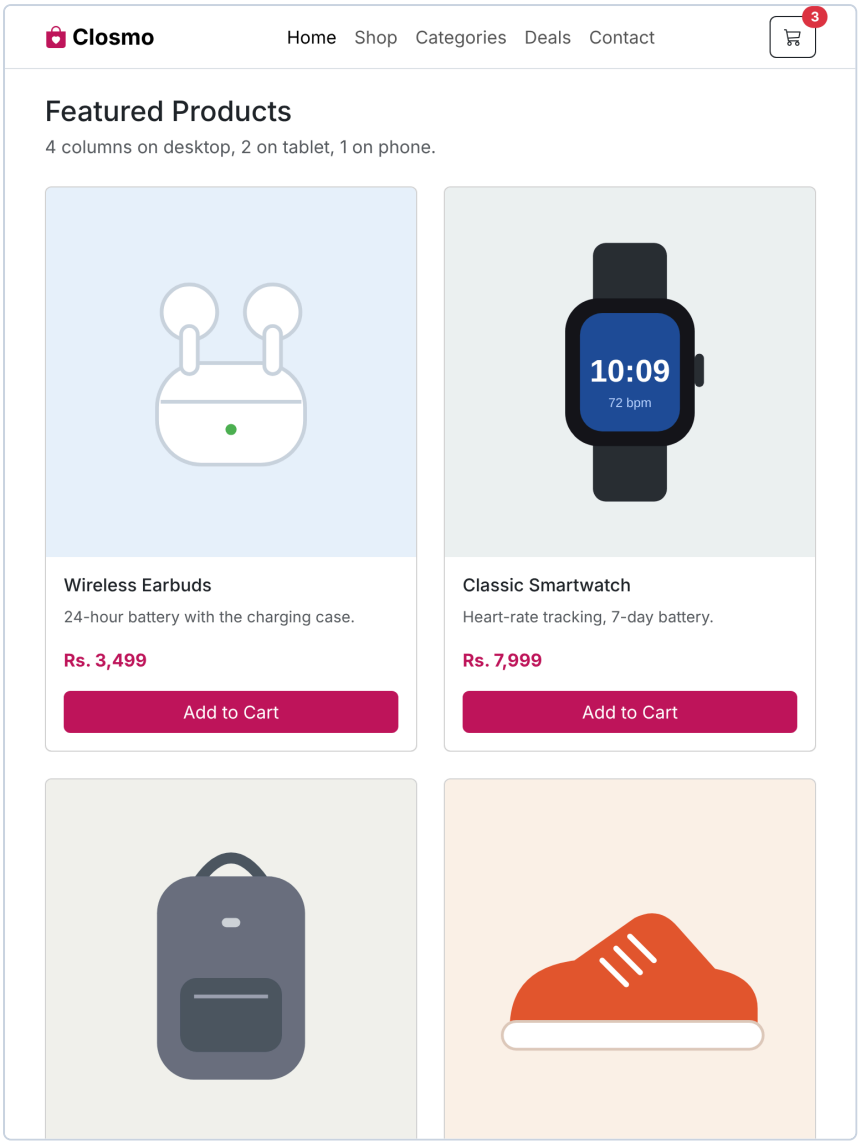
2. Screenshots of the finished work running in a browser

Desktop — 1280px (4 columns, links centred, cart right)



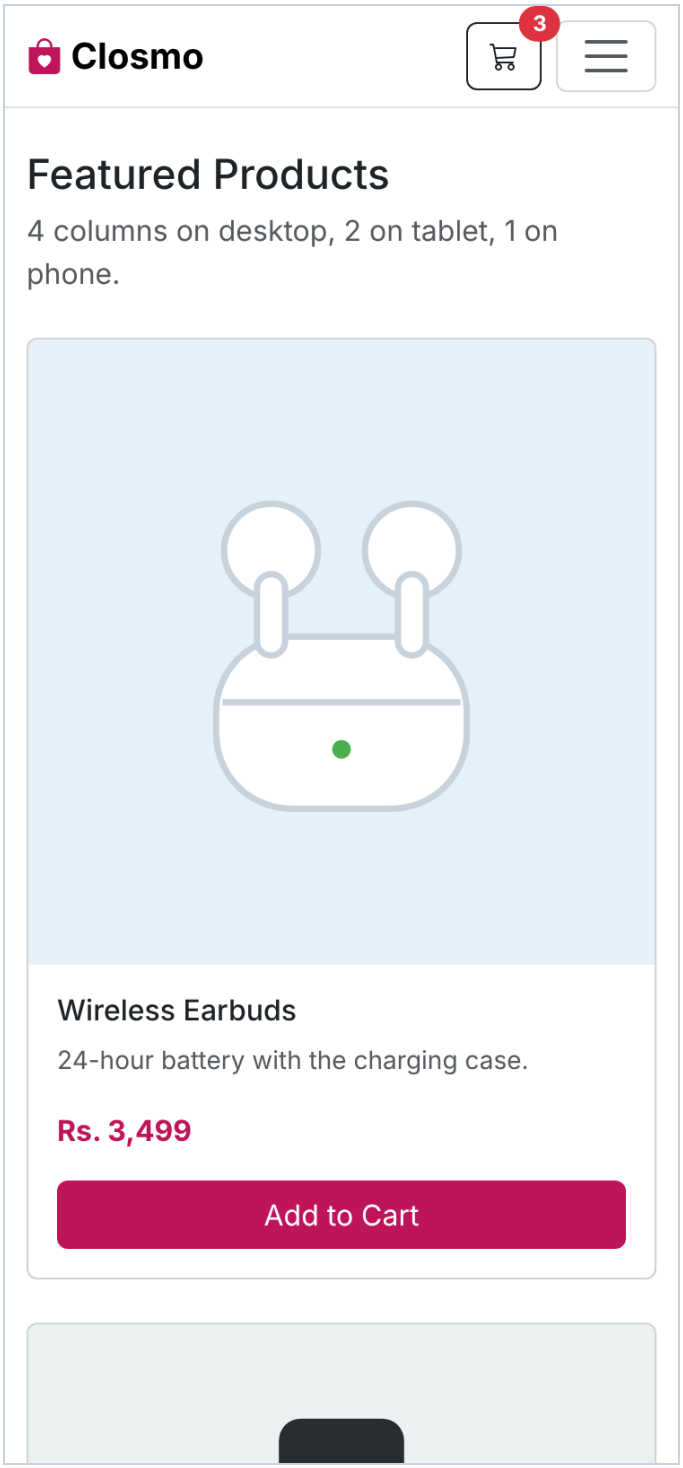
Full page at 1280px: `col-lg-3` gives 4 cards per row; the navbar is expanded.

Tablet — 768px (2 columns)

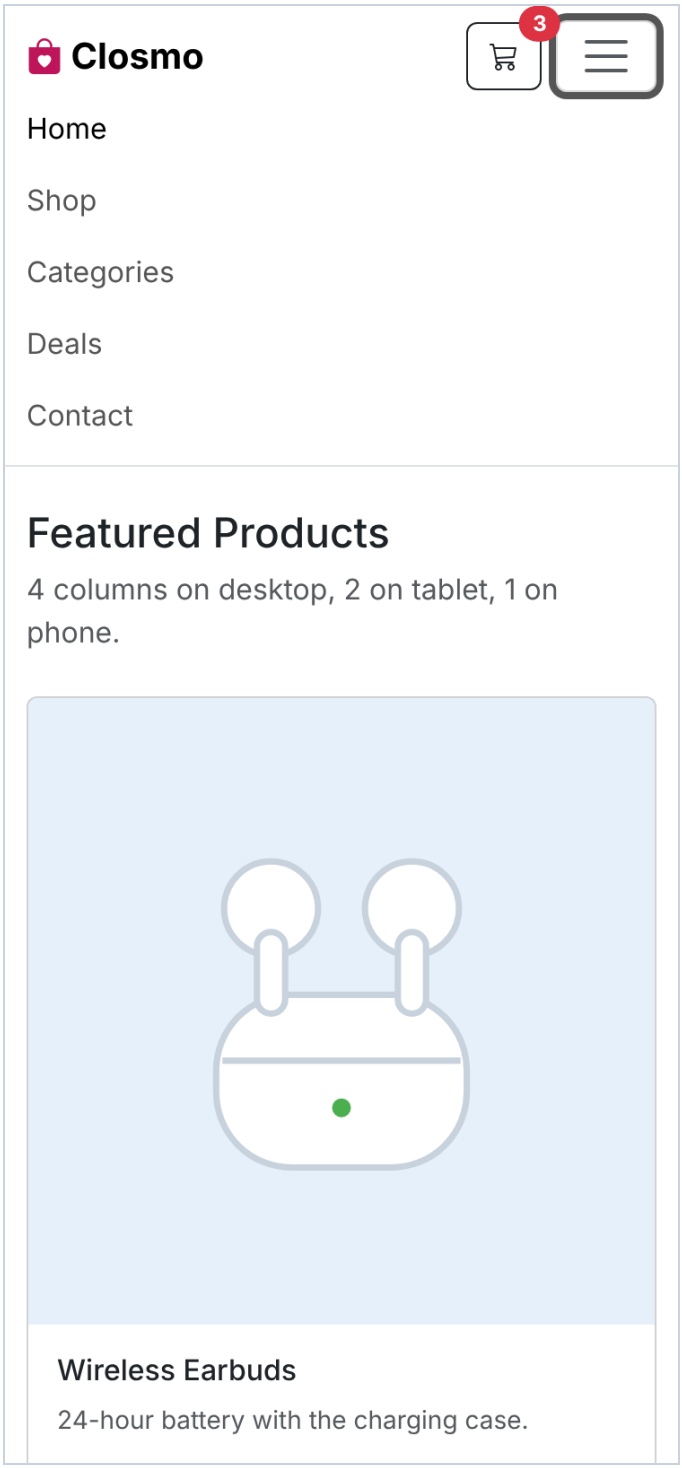


768 × 1024: `col-md-6` gives 2 cards per row; 768px is the `md` breakpoint, so the links are still shown in a row.

Phone — 375px (1 column, hamburger)



375 × 812, menu closed: links hidden behind the hamburger; `col-12` gives 1 card per row.



375 × 812, after tapping the hamburger: Bootstrap's collapse shows the links.

3. Code

Framework files: Bootstrap 5.3.3 CSS and JS bundle and Bootstrap Icons 1.11.3, loaded from the jsDelivr CDN (see the `<link>` and `<script>` tags). Images: 8 SVG files in `images/`.

lab-2/index.html

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
    <meta charset="UTF-8">
    <!-- Without this tag phones render a fake ~980px page and zoom out,
         so the Bootstrap breakpoints (media queries) would never fire. -->
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Closmo | Responsive Navbar and Grid</title>

    <!-- Bootstrap 5.3.3 CSS from the jsDelivr CDN -->
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css">
    <!-- Bootstrap Icons: an icon font, used for the logo and the cart -->
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-
icons.min.css">
    <!-- My own CSS loads AFTER Bootstrap so my rules win ties (same specificity, later source order) -->
    <link rel="stylesheet" href="styles.css">
</head>
<body>

    <header>
        <!-- navbar-expand-md: links are shown in a row from the md breakpoint (768px) up;
             below 768px they collapse behind the hamburger button. -->
        <nav class="navbar navbar-expand-md bg-white border-bottom sticky-top" aria-label="Main navigation">
            <div class="container">

                <!-- 1. Logo (left) -->
                <a class="navbar-brand fw-bold" href="index.html">
                    <i class="bi bi-bag-heart-fill text-brand" aria-hidden="true"></i> Closmo
                </a>

                <!-- 3. Cart + hamburger (right). order-md-last moves this group after the
                     links on tablets and desktops, while on phones it stays on the first row
                     next to the logo, so the cart is always visible. -->
                <div class="nav-actions d-flex align-items-center gap-2 order-md-last">
                    <a href="#" class="btn btn-outline-dark position-relative" aria-label="Cart, 3 items">
                        <i class="bi bi-cart3" aria-hidden="true"></i>
                        <span class="position-absolute top-0 start-100 translate-middle badge rounded-pill
bg-danger">3</span>
                    </a>
                    <!-- The hamburger. Bootstrap's own JS (bundle below) reads data-bs-toggle and
                         data-bs-target and shows/hides #mainNav. navbar-expand-md hides this
                         button automatically from 768px up. -->
                    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#mainNav"
                            aria-controls="mainNav" aria-expanded="false" aria-label="Toggle navigation">
                        <span class="navbar-toggler-icon"></span>
                    </button>
                </div>

                <!-- 2. Links (centre). The collapse div fills the space between the logo and the cart;
                     mx-auto (margin-left and margin-right: auto) centres the list inside that space. -->
                <div class="collapse navbar-collapse" id="mainNav">
                    <ul class="navbar-nav mx-auto">
                        <li class="nav-item"><a class="nav-link active" aria-current="page"
href="index.html">Home</a></li>
                        <li class="nav-item"><a class="nav-link" href="#products">Shop</a></li>
                        <li class="nav-item"><a class="nav-link" href="#">Categories</a></li>
                        <li class="nav-item"><a class="nav-link" href="#">Deals</a></li>
                        <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
                    </ul>
                </div>

            </div>
        </nav>
    </header>

    <main class="container py-4" id="products">
        <h1 class="h3 mb-1">Featured Products</h1>
        <p class="text-muted mb-4">4 columns on desktop, 2 on tablet, 1 on phone.</p>

        <!-- Bootstrap grid: .row is a flex container; each column below is a flex item.
             A row has 12 units, so:
               col-12    -> 12/12 = 1 card per row  (phones, below 768px)
```

```
               col-md-6  -> 12/6  = 2 cards per row (tablets, 768px and up)
               col-lg-3  -> 12/3  = 4 cards per row (desktops, 992px and up)
             g-4 sets the gutter (gap) between columns and rows. -->
        <div class="row g-4">

            <div class="col-12 col-md-6 col-lg-3">
                <article class="card h-100 product-card">
                    <img src="images/earbuds.svg" class="card-img-top" alt="Wireless earbuds in a charging
case">
                    <div class="card-body d-flex flex-column">
                        <h2 class="card-title h6">Wireless Earbuds</h2>
                        <p class="card-text small text-muted">24-hour battery with the charging case.</p>
                        <p class="price">Rs. 3,499</p>
                        <button type="button" class="btn btn-brand mt-auto">Add to Cart</button>
                    </div>
                </article>
            </div>

            <div class="col-12 col-md-6 col-lg-3">
                <article class="card h-100 product-card">
                    <img src="images/smartwatch.svg" class="card-img-top" alt="Black smartwatch with a blue
screen">
                    <div class="card-body d-flex flex-column">
                        <h2 class="card-title h6">Classic Smartwatch</h2>
                        <p class="card-text small text-muted">Heart-rate tracking, 7-day battery.</p>
                        <p class="price">Rs. 7,999</p>
                        <button type="button" class="btn btn-brand mt-auto">Add to Cart</button>
                    </div>
                </article>
            </div>

            <div class="col-12 col-md-6 col-lg-3">
                <article class="card h-100 product-card">
                    <img src="images/backpack.svg" class="card-img-top" alt="Grey backpack with a front
pocket">
                    <div class="card-body d-flex flex-column">
                        <h2 class="card-title h6">Everyday Backpack</h2>
                        <p class="card-text small text-muted">Water-resistant, fits a 15.6-inch laptop.</p>
                        <p class="price">Rs. 4,250</p>
                        <button type="button" class="btn btn-brand mt-auto">Add to Cart</button>
                    </div>
                </article>
            </div>

            <div class="col-12 col-md-6 col-lg-3">
                <article class="card h-100 product-card">
                    <img src="images/shoes.svg" class="card-img-top" alt="Orange running shoe">
                    <div class="card-body d-flex flex-column">
                        <h2 class="card-title h6">Running Shoes</h2>
                        <p class="card-text small text-muted">Light mesh upper, cushioned sole.</p>
                        <p class="price">Rs. 5,600</p>
                        <button type="button" class="btn btn-brand mt-auto">Add to Cart</button>
                    </div>
                </article>
            </div>

            <div class="col-12 col-md-6 col-lg-3">
                <article class="card h-100 product-card">
                    <img src="images/lamp.svg" class="card-img-top" alt="Yellow desk lamp">
                    <div class="card-body d-flex flex-column">
                        <h2 class="card-title h6">Study Desk Lamp</h2>
                        <p class="card-text small text-muted">Adjustable arm, warm LED light.</p>
                        <p class="price">Rs. 2,150</p>
                        <button type="button" class="btn btn-brand mt-auto">Add to Cart</button>
                    </div>
                </article>
            </div>

            <div class="col-12 col-md-6 col-lg-3">
                <article class="card h-100 product-card">
                    <img src="images/mug.svg" class="card-img-top" alt="Green ceramic mug">
```

```
                    <div class="card-body d-flex flex-column">
                        <h2 class="card-title h6">Ceramic Mug</h2>
                        <p class="card-text small text-muted">350 ml, dishwasher safe.</p>
                        <p class="price">Rs. 899</p>
                        <button type="button" class="btn btn-brand mt-auto">Add to Cart</button>
                    </div>
                </article>
            </div>

            <div class="col-12 col-md-6 col-lg-3">
                <article class="card h-100 product-card">
                    <img src="images/powerbank.svg" class="card-img-top" alt="Black 20000 mAh power bank">
                    <div class="card-body d-flex flex-column">
                        <h2 class="card-title h6">Power Bank 20000 mAh</h2>
                        <p class="card-text small text-muted">Charges two phones at once.</p>
                        <p class="price">Rs. 4,999</p>
                        <button type="button" class="btn btn-brand mt-auto">Add to Cart</button>
                    </div>
                </article>
            </div>

            <div class="col-12 col-md-6 col-lg-3">
                <article class="card h-100 product-card">
                    <img src="images/sunglasses.svg" class="card-img-top" alt="Black sunglasses">
                    <div class="card-body d-flex flex-column">
                        <h2 class="card-title h6">Polarized Sunglasses</h2>
                        <p class="card-text small text-muted">UV400 lenses, lightweight frame.</p>
                        <p class="price">Rs. 1,799</p>
                        <button type="button" class="btn btn-brand mt-auto">Add to Cart</button>
                    </div>
                </article>
            </div>

        </div>
    </main>

    <!-- Bootstrap's JS bundle (includes Popper). Only needed here for the hamburger collapse.
         Loaded at the end of <body> so the HTML above already exists when it runs. -->
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

lab-2/styles.css

```
/* Lab 2 - my own CSS on top of Bootstrap.
   This file is linked after bootstrap.min.css, so if two rules are equally
   specific, mine wins (the later rule wins). */

/* Brand colours, defined once here and reused with var() */
:root {
    --brand: #c2185b;
    --brand-dark: #8e0e41;
}

/* Logo icon colour */
.text-brand {
    color: var(--brand);
}

/* Add to Cart button colour (Bootstrap's .btn gives the shape, this gives the colour) */
.btn-brand {
    background-color: var(--brand);
    border-color: var(--brand);
    color: #ffffff;
}

.btn-brand:hover {
    background-color: var(--brand-dark);
    border-color: var(--brand-dark);
    color: #ffffff;
```

```
}

/* Product cards lift a little on hover */
.product-card {
    transition: transform 0.2s ease, box-shadow 0.2s ease;
}

.product-card:hover {
    transform: translateY(-4px);
    box-shadow: 0 8px 20px rgba(0, 0, 0, 0.12);
}

/* Square images, so every card is the same height */
.product-card .card-img-top {
    aspect-ratio: 1 / 1;
    object-fit: cover;
}

.price {
    font-weight: 700;
    color: var(--brand);
}
```